

The nautilus workflow

Two input streams and a parallel video workflow converge into the core processing pipeline before branching into specialized, optional analyses.

Deployment Metadata

one row per deployment: ID, dates, coordinates, tag model, axis config, paddle wheel, biological traits

Primary Tag Data

CATS / **CEiiA** multi-sensor CSV
Little Leonardo .txt loggers

Co-deployed Satellite Tag

optional second input stream
Wildlife Computers **MiniPAT** / **MK10** / **SPOT**
GPS & Argos fixes, wet / dry

Onboard Video

optional · camera tags only
raw clips .mp4 / .mov

1 · Prepare & Import

- `checkDeploymentMetadata()`
- `importTagData()` ---> *cats / little_leonardo*
- `buildTagData()` ---> *when the data is already in R*

validates deployment metadata prior to import, standardizes sensor data, and merges co-deployed spatial and wet/dry records

2 · Clean & Quality-Control

- `filterDeploymentData()`
- `regularizeTimeSeries()`
- `filterLocations()`
- `checkSensorIntegrity()`
- `checkSensorQuality()`

trims data to the deployment period, regularises sampling, and performs automated quality checks on sensor and position data

3 · Orient & Calibrate

- `checkTagMapping()`
- `consensusAxisMapping()`
- `calibrateMagnetometer()`
- `reviewTagMapping()` *optional: video-assisted verification of tag mapping*
- `applyAxisMapping()`

determines tag orientation by validating documented configurations against sensor data and video

Video Utilities

- `reencodeVideos()`
compresses raw footage to HEVC format
- `installCamOcrModel()`
one-off: optical character recognition (OCR) initialization for burned-in timestamps.
- `getVideoMetadata()`
extracts clip start, duration, and frame rate from filenames or via OCR
- `saveUncertainTimestampFrames()`
facilitates manual visual confirmation of uncertain timestamps
- `launchVideo()`
enables direct navigation to specific video events

synchronizes video and sensor timestamps to facilitate precise visual axis review

analysis-ready
nautilus_tag

processTagData()

VeDBA, ODBA, jerk, roll, pitch, heading, turning angle, surge, sway, heave, paddle-wheel freq, vertical velocity

optional branches

Summaries & Figures

per animal and cohort

- `summarizeTagData()`
- `plotDepthProfiles()`
- `plotTimeAtDepth()`
- `plotDistributions()`

aggregates data summaries per animal or cohort and standardizes visualization aesthetics globally via `plotTheme()`

Dive Analysis

one row per dive

- `detectDives()`
- `diveMetrics()`
- `plotDives()`

two-threshold dive detection using a fixed reference or adaptive baseline

Behaviour & Kinematics

swimming effort, ML features

- `calculatePaddleSpeed()`
- `calculateTailBeats()`
- `extractFeatures()`
- `getDielPhase()`

computes rolling-window kinematics and temporal features optimized for behavioral modeling

Movement Tracks

dead-reckon and validate

- `reconstructTrack()`
- `crossValidateTrack()`
- `trackMetrics()`
- `plotTracks()`
- `exportForSSM()`

reconstructs, validates, and exports dead-reckoned trajectories for state-space modeling

Video & Annotation

score behaviour onto samples

- `filterVideoPeriod()`
- `annotateData()`
- `renderOverlayVideo()`

annotates behaviors for classifier training sets and generates data-overlaid video outputs

Package Utilities & Diagnostics

- `getTagMetadata()`
- `updateTagMetadata()`
- `processingHistory()`
- `processingSummary()`
- `issues()`

retrieves or updates object metadata — each nautilus object maintains embedded traits and a reproducible, append-only processing audit trail